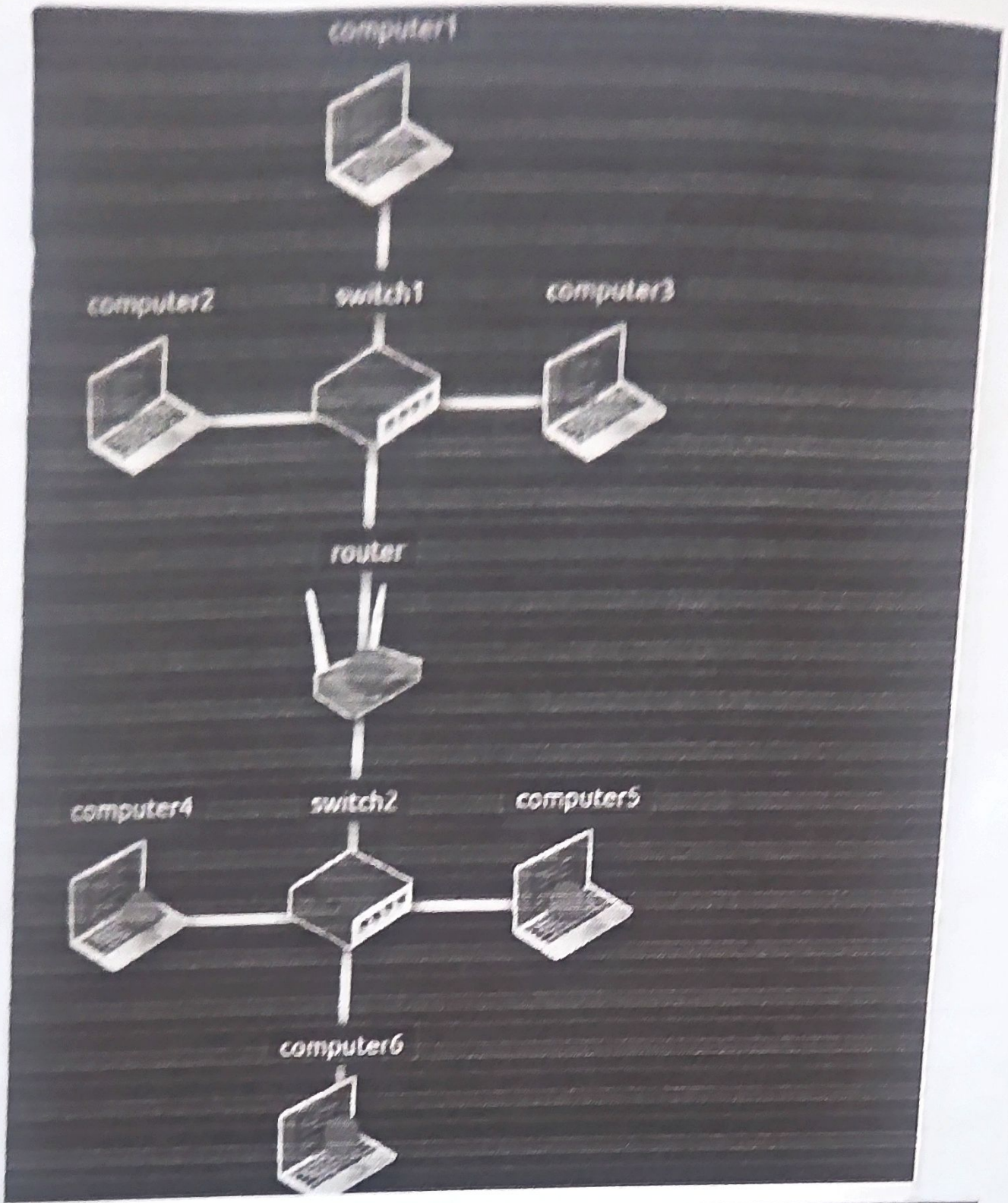
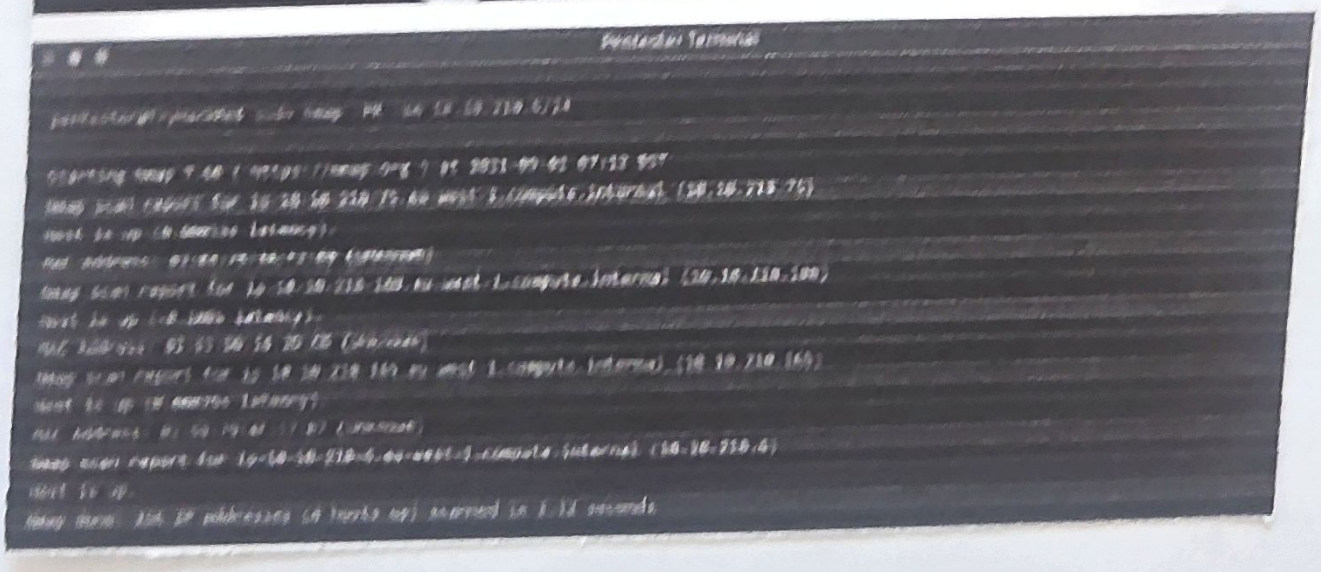

computer1
computer2
switch1
computer3
router
computer4
switch2
computer5
computer6



Exp - 8 NMAP to discover Live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the Tryhackme platform.

AIM:

To discover Live Hosts using Nmap Scans (ARP, ICMP, TCP/LRP) on the Tryhackme platform room link https://tryhackme.com/room/nmap01

1. Introduction to nmap.

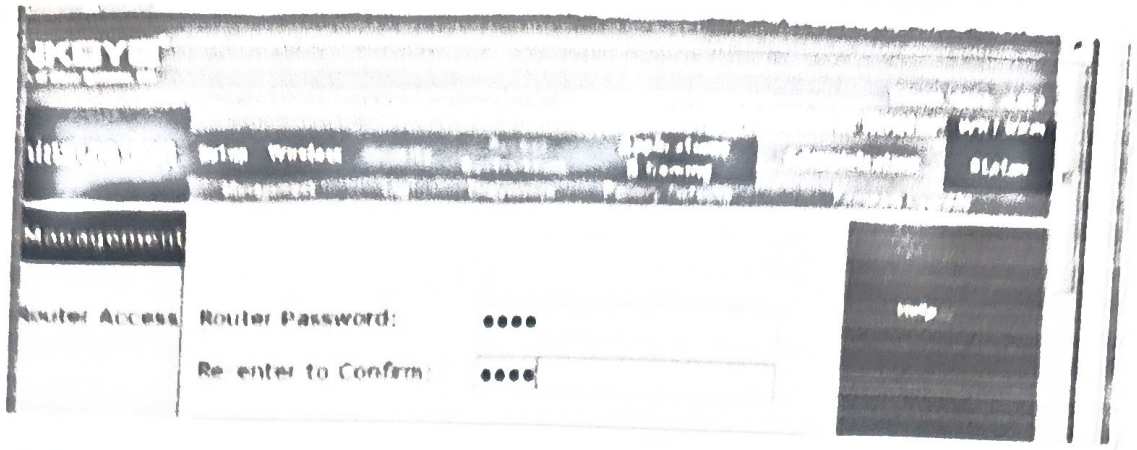
nmap is a free and open-source network scanning tool used to discover active systems and Identify running services on a network before performing port scanning It is useful to determine which systems are actualy online scanning offline systems wastes time and can create unnecessary network traffic Nmap provides different methods for live host directionary including.

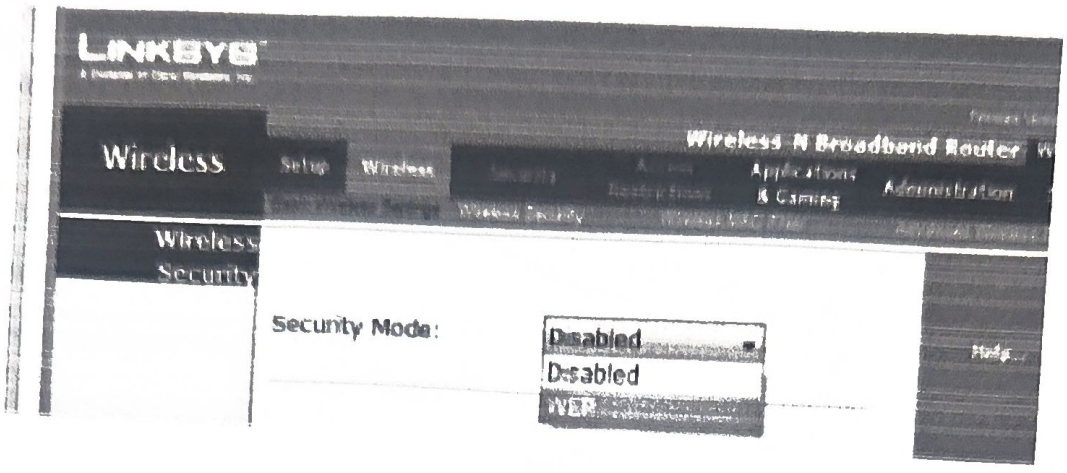
- ARP Scan
- Icmp Scan
- TCP SYN Ping
- TCP ACK Ping

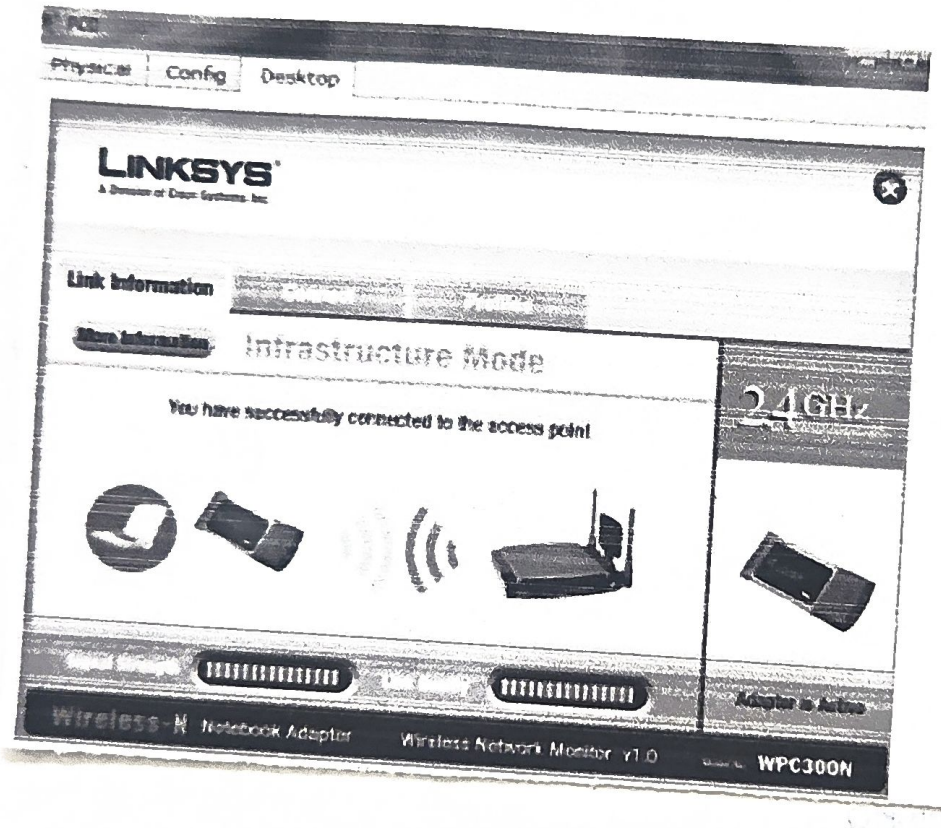
The exp is based on the Nmap live host Discovery room of Try-Hack-me.

2. Subnetworks

A network segment is a group of computer connected through a common medium such as an Ethernet switch/ WIFI access point a subnetwork (subnet) is a logical Portion of an IP network having its own IP address range.


Management
Router Access
Router Password:
Re-enter to Confirm:
Help


LINKSYS
Wireless
Setup
Wireless
Wireless N Broadband Router
Applications & Gaming
Administration
Wireless Security
Security Mode:
Disabled
Disabled
Help...


Physical
Config
Desktop
LINKSYS
Link Information
Intrastructure Mode
You have successfully connected to the access point
2.4GHz
Wireless-N Notebook Adapter
Wireless Network Monitor V1.0
WPC300N

Ex

Subnet	Subnet mask	Approx hosts
/16	255.255.0.0	~ 65,000
/24	255.255.255.0	~ 250

ARP - based discovery works only within the same subnet because ARP operates at the link layer and its requests cannot cross router.

3. Enumerating target

Before scanning Nmap needs to know which systems/IP addresses should be checked.

Target can be specified as :

→ Single IP address : 10.10.105

→ Range of IP address : 10.10.10.5-10

→ Subnet : 10.10.10.0/24

→ List of Targets from a file: map-il lists-of-hosts-txt

This target enumeration helps define the scope of the scan

4. Live host discovery

The experiment covers 4 protocols

ARP - Link layer

ICMP - network layer

TCP - Trap out layer

UDP - Transport layer.

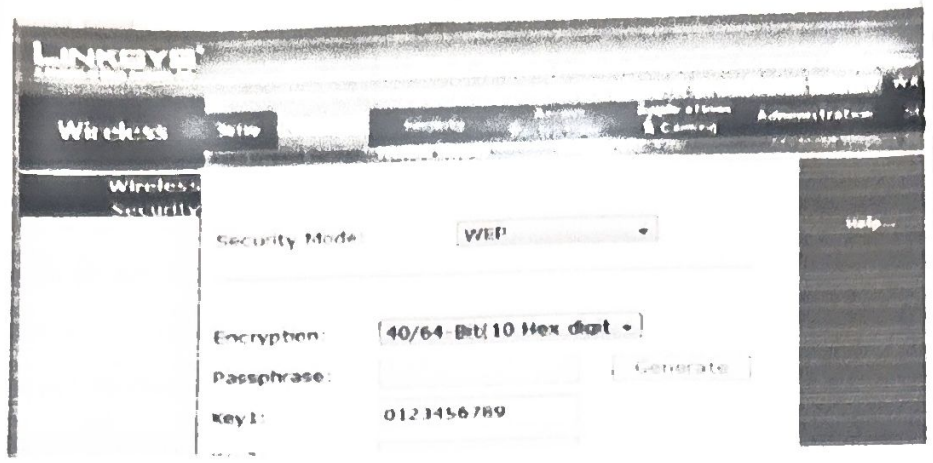
5. ARP host Discovery:

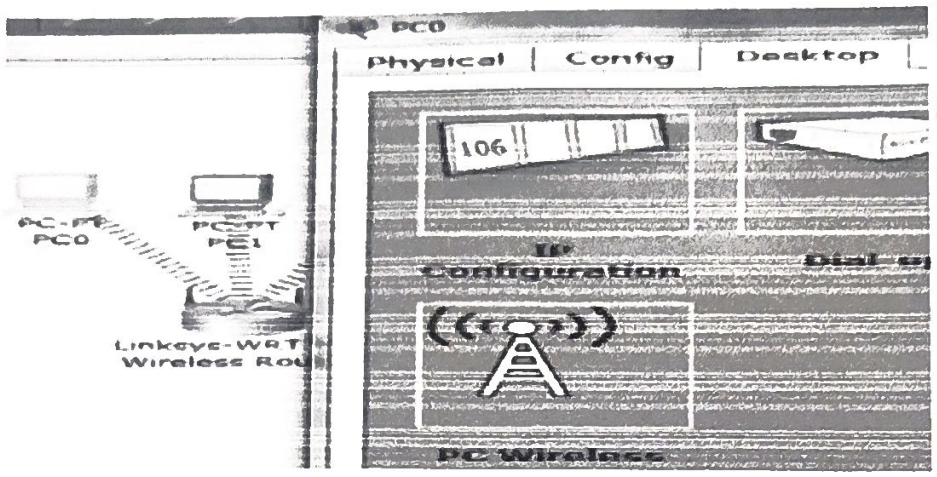
→ ARP requests are used to find live hosts on the same subnet.

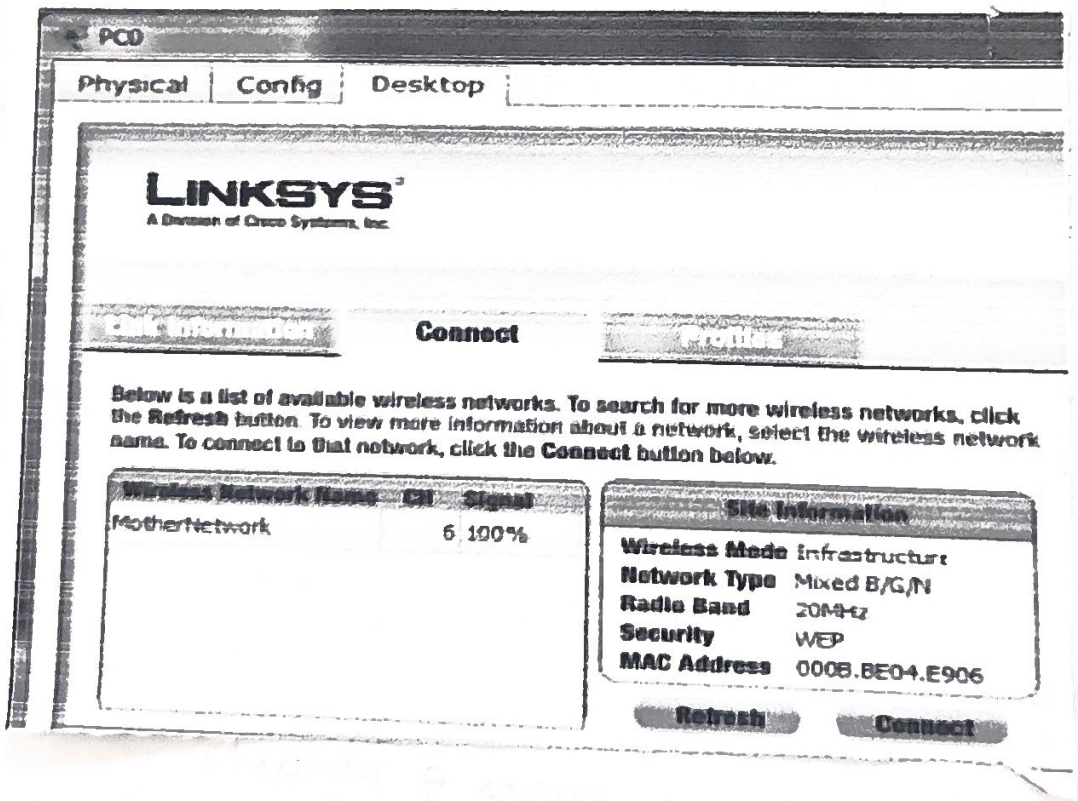
→ A live host responds with an ARP reply.

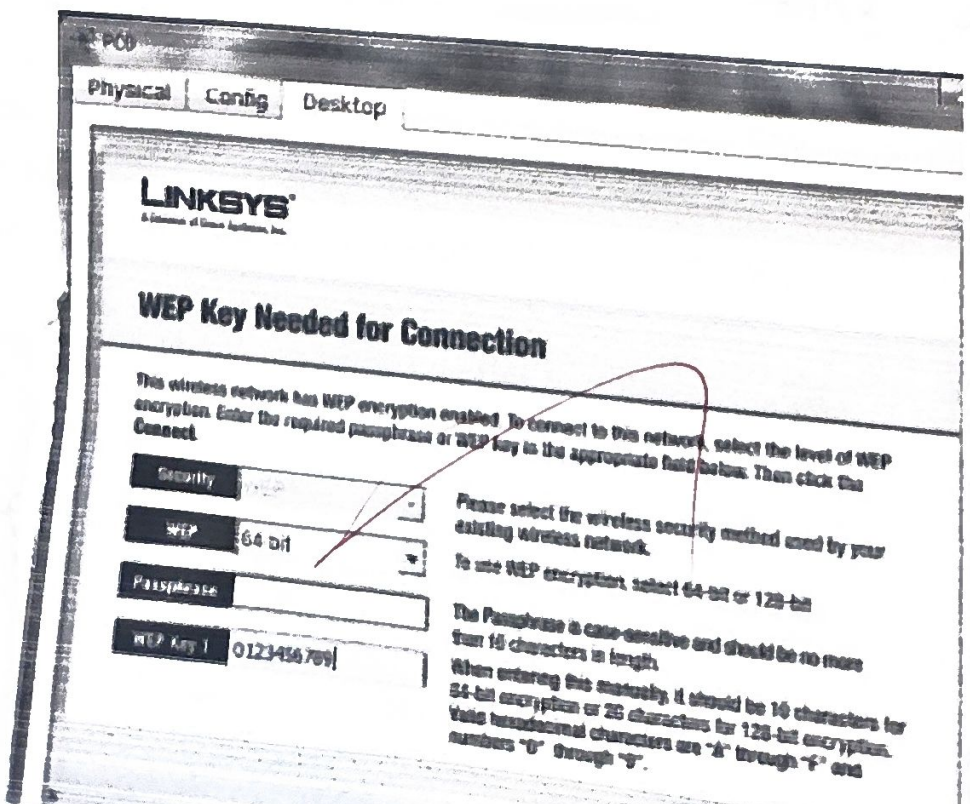
→ ARP cannot cross router

→ PR -sn Targets platform an ARP based









6. ICMP [illegible]
   -> RCMP [illegible]
   (Echo request)
   -> Nmap [illegible]
   -> Rcmp Timestamp [illegible]
   -> Icmp address mask
   -> Rcmp may not [illegible]

7 TCP SYN ping
   -> Sends a TCP packet
   -> An open port may
   -> A closed port
   -> Nmap options -PS

8. TCP ACK ping.
   -> Sends a TCP packet
   -> Normally used
   -> Nmap option: -
   -> A Target may
   the host is active.

9. UDP Ping.
   -> UDP packets ca
   host in active.
   -> a packet se
   an RCMP port
   -> This respon
   active and

10. Reverse DNS
   -> Nmap
   for other h

6. ICMP host discovery

-> ICMP echo uses Type 8 (Echo Request) and Type 0 (Echo reply)

-> Nmap option: - PE

-> ICMP Timestamp scan - PP

-> Icmp address mask scan: - Pm

-> Icmp may not always work because firewalls can

7 TCP SYN Ping

-> Sends a TCP packet with SYN flag set

-> An open port may respond with SYN/ACK

-> A closed port may respond with RST

-> Nmap options -PS

8. TCP ACK ping.

-> Sends a TCP packet with the ACK flag set.

-> Normally uses port 80 by default.

-> Nmap option: -PA.

-> A Target may respond with a RST, Indicating that the host is active.

9. UDP Ping.

-> UDP packets can be used to determine whether a host in active.

-> a packet sent to a closed UDP port may produce an ICMP port unreachable respond.

-> This response indicates that the target host is active and reachable.

10. Reverse DNS lookup.

-> Nmap normally performs reverse - DNS lookups for other hosts.

-> -n -> disables DNS resolution

-> -R - performs reverse-DNS lookup even for off line hosts

Purpose	Option
PiY / host discover only.	-sN
ARP scan	-PR
ICMP Echo	-PE
ICMP Timestamp	-PP
Icmp address mask	-PM
ICP SYP Ping	-PS
TCP Ack ping	-PA
Disable DNS Lookup	-n
Force reverse DNS	-R
	-SL

Result:

The Nmap live host Discovery experiment was successfully studied on the Try Hack me platform hive hosts some identified ARP, ICMP, TCP and UDP techniques and the corresponding nmapp often were understand.